

Transactions:

- ☐ Series of operations
- ☐ all the operations inside a transaction must be completed; otherwise, it will roll back to the previous state

for example, online banking, where transferring money from one account to another involves debiting the amount in one account and simultaneously crediting that amount in the other.

Start of a Transaction:

```
START TRANSACTION;
```

Execute SQL Statements:

updates, insertions, or deletions of records.

```
UPDATE accounts SET balance = balance - 100 WHERE account_id = 1;
```

```
UPDATE accounts SET balance = balance + 100 WHERE account_id = 2;
```

Committing a Transaction:

makes all of the changes permanent and terminates the transaction.

```
COMMIT;
```

Rolling Back a Transaction:

- If something goes wrong and you want to "roll back" the changes made during the transaction,

```
ROLLBACK;
```

COMPLETE TRANSACTION:

Create Table

```
CREATE TABLE accounts (  
    account_id INT PRIMARY KEY,  
    account_name VARCHAR(100),  
    balance DECIMAL(10, 2)  
);
```

Insert Data

```
INSERT INTO accounts (account_id, account_name, balance) VALUES  
(1, 'Alice', 1000.00),  
(2, 'Bob', 500.00);
```

```
START TRANSACTION;
```

```
UPDATE accounts SET balance = balance - 100 WHERE account_id = 1;
```

```
UPDATE accounts SET balance = balance + 100 WHERE account_id = 2;
```

```
COMMIT;  saves the transaction
```

Output :

account_id	account_name	balance
1	Alice	900.00
2	Bob	600.00

Using Savepoints

intermediate points within a transaction

partial transaction rollbacks without affecting whole transaction

Start Transaction

```
START TRANSACTION;
```

```
SAVEPOINT savepoint1;
```

```
UPDATE accounts SET balance = balance - 100 WHERE account_id = 1;
```

```
SAVEPOINT savepoint2;
```

```
UPDATE accounts SET balance = balance + 100 WHERE account_id = 2;
```

```
-- If an error occurs, roll back to a specific savepoint
```

```
ROLLBACK TO SAVEPOINT savepoint1;
```

```
-- Finally, commit the transaction
```

```
COMMIT;
```

Output:

account_id	account_name	balance
1	Alice	1000.00
2	Bob	500.00